CLINICAL · EAR

Intracranial complications of otitis media

A discharging ear with headache, fever, fits or drowsiness is an intracranial complication until proven otherwise. Meningitis is the commonest, brain abscess the next; often there is more than one. The order of work matters: resuscitate, image before any lumbar puncture, start antibiotics, involve neurosurgery, and only then deal with the ear.

AT A GLANCE

- Discharging ear plus fever, headache, seizures, focal signs or drowsiness.
- Meningitis is commonest; brain abscess is next.
- Contrast imaging comes before any lumbar puncture.
- IV antibiotics, neurosurgery, then treat the ear source.
- Check hearing afterwards.

1 CHAPTER ONE Overview

When otitis media crosses the dura.

1.1 Definition and scope

Infection spreading from the middle ear and mastoid into the cranial cavity: extradural abscess, subdural empyema, meningitis, brain abscess, lateral (sigmoid) sinus thrombophlebitis and otitic hydrocephalus. Perforation, ossicular erosion, tympanosclerosis and atelectasis are *sequelae* of otitis media, not complications.

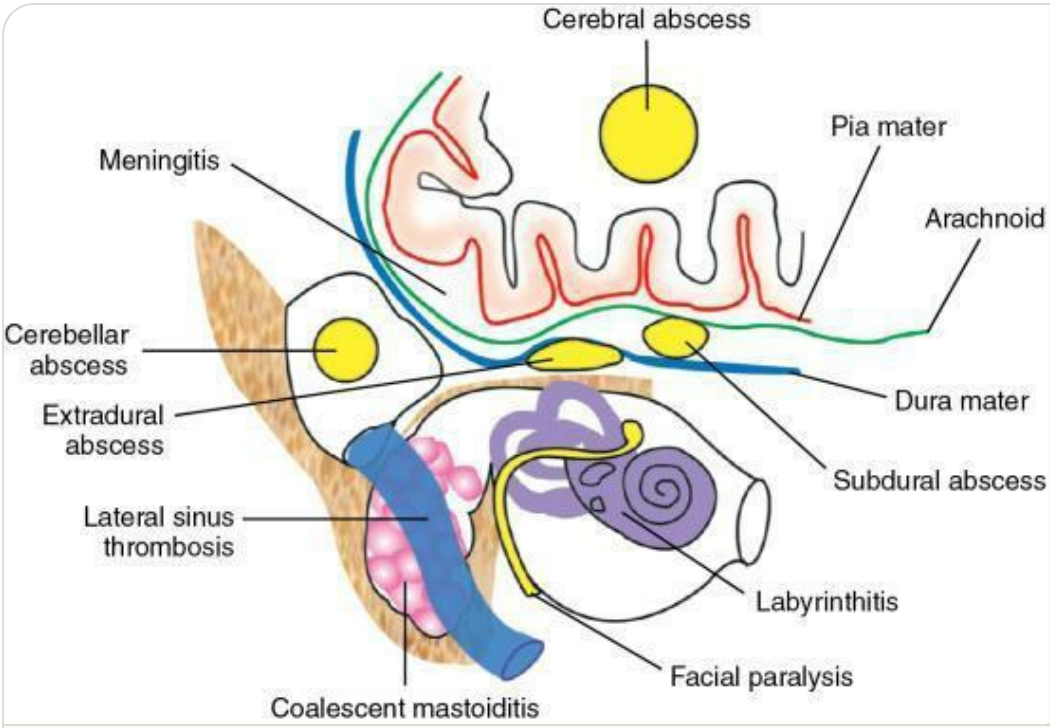


FIGURE 1 The complications of suppurative otitis media and the routes of spread.
Dhingra, fig. 12.1

1.2 Epidemiology

Antibiotics have made these complications rare where care is good, but they remain important where chronic ears go untreated. Half of brain abscesses in adults and a quarter in children are otogenic, and a temporal lobe abscess is about twice as common as a cerebellar one.

1.3 Acute or chronic disease

Infants with acute otitis media tend to develop meningitis by blood-borne spread. Adults with chronic otitis media, especially cholesteatoma, develop abscesses and sinus thrombosis by bone erosion.

WHERE THE BOOKS DIFFER · HOW OFTEN THERE IS MORE THAN ONE

Dhingra	Lee
About one-third of patients with otogenic meningitis have another intracranial lesion.	About half.

Either way, look for a second complication in every patient.

2 CHAPTER TWO Pathogenesis

Three routes into the skull.

2.1 Routes of spread

2.1.1 Direct bone erosion

Cholesteatoma, osteitis and granulation destroy the tegmen or the sinus plate, bringing infection to the dura.

2.1.2 Venous thrombophlebitis

Infected thrombus in small veins crosses intact bone, which is why an abscess can arise without visible erosion.

2.1.3 Preformed pathways

Congenital dehiscences, a patent petrosquamous suture or Hyrtl's fissure in children, inner ear malformations, fractures, previous surgery, and the round and oval windows via the labyrinth.

2.2 Risk factors

Extremes of age, poverty and poor access to care, virulent or resistant organisms, inadequate antibiotic courses, immunosuppression, preformed defects and cholesteatoma.

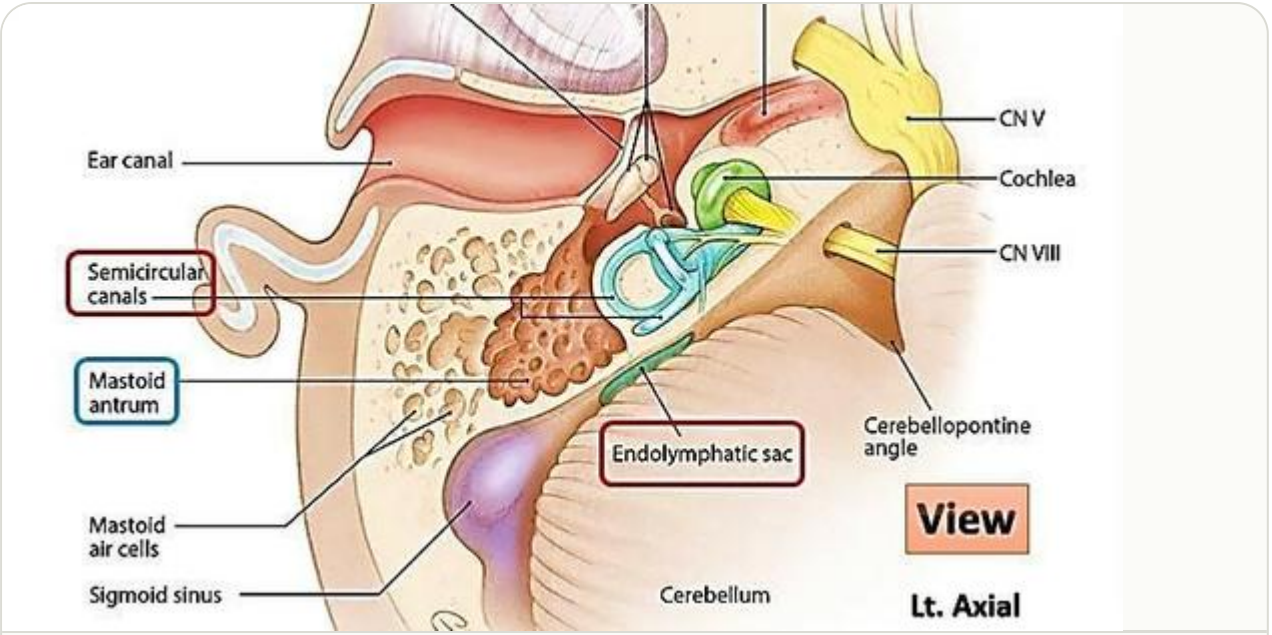


FIGURE 2 Axial relations of the middle ear and mastoid: the dura, sigmoid sinus and labyrinth that infection can reach.

Anatomy atlas · ENT Steps p. 57

2.3 Organisms

Mixed: *Staphylococcus aureus*, streptococci, *Proteus*, *Pseudomonas* and *E. coli*, with anaerobes such as *Bacteroides*, particularly in abscesses.

3 CHAPTER THREE The complications

Each with its own picture.

3.1 Extradural abscess

3.1.1 Picture

Pus between the bone and the dura of the middle or posterior fossa. Often silent and found at mastoidectomy; sometimes a deep, one-sided headache that eases when pus flows from the ear.

3.1.2 Treatment

At mastoidectomy, remove bone until healthy dura is exposed. If pus is suspected behind an intact plate, take the plate down.

3.2 Subdural empyema

3.2.1 Picture

Pus in the subdural space spreads quickly over the hemisphere: severe headache, fever, meningism, seizures, aphasia and rapidly progressive hemiplegia, with raised pressure.

3.2.2 Treatment

A neurosurgical emergency: urgent drainage through burr holes or craniotomy, with high-dose antibiotics.

3.3 Meningitis

3.3.1 Picture

The commonest complication. Headache, fever, photophobia, vomiting, neck stiffness, and positive Kernig and Brudzinski signs; later, drowsiness, fits and coma.

3.3.2 CSF

Turbid, under raised pressure, with many polymorphs, high protein and low sugar.

3.3.3 Treatment

Intravenous antibiotics that penetrate the CSF, and steroids. Treat the ear: myringotomy for acute otitis media, mastoidectomy for chronic disease once the patient is fit.

AFTER MENINGITIS

Check hearing early. Labyrinthitis ossificans can close the cochlea within weeks, and a cochlear implant must go in before it does.

3.4 Brain abscess

3.4.1 Stages

WHERE THE BOOKS DIFFER · STAGES OF AN OTOGENIC BRAIN ABSCESS

Dhingra	Lee
Four stages: invasion (initial encephalitis), localisation (the latent, quiet stage), enlargement, and termination by rupture into the ventricle or subarachnoid space.	Three stages: encephalitis, a quiet period, and rupture.

The quiet stage is the trap: the patient seems better while the abscess is forming.

3.4.2 Raised intracranial pressure

Morning headache, projectile vomiting, papilloedema, a slow pulse and a subnormal temperature, with drowsiness.

3.4.3 Temporal lobe abscess

Nominal aphasia (with a dominant-side lesion), a contralateral upper quadrant field defect, contralateral weakness, and uncinate fits with hallucinations of taste and smell. An ipsilateral third nerve palsy with a dilated pupil warns of tentorial herniation.

3.4.4 Cerebellar abscess

Suboccipital headache, nystagmus, ipsilateral ataxia with past-pointing, intention tremor and dysdiadochokinesia, and early papilloedema.

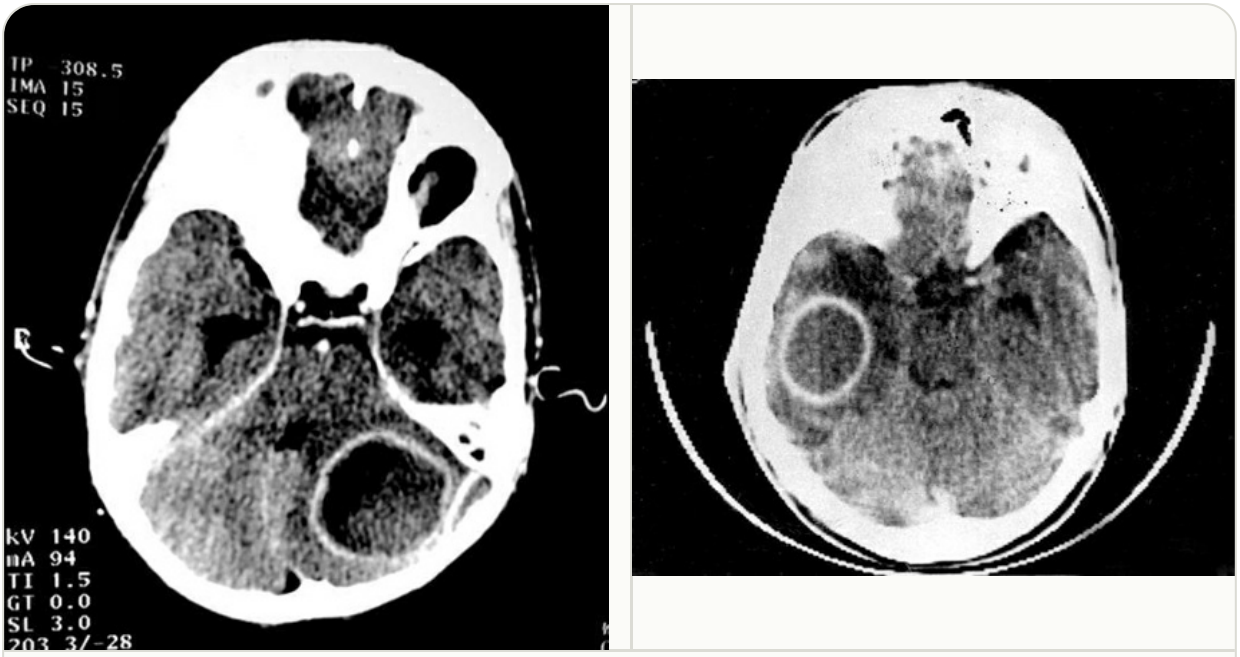


FIGURE 3 First, CT of a left cerebellar abscess. Second, CT of a right otogenic cerebral (temporal lobe) abscess.

Dhingra, figs 12.6 and 12.7

3.4.5 CSF

Moderately raised cells and protein with normal sugar; but a lumbar puncture is dangerous and is avoided while there is mass effect.

3.4.6 Treatment

Broad-spectrum intravenous antibiotics with anaerobic cover, steroids or mannitol for raised pressure, and neurosurgical drainage by aspiration through a burr hole or excision. The ear is then made safe through a separate operative field.

WHERE THE BOOKS DIFFER · TIMING OF EAR SURGERY WITH AN ABSCESS

Dhingra	Lee
Operate on the ear once the abscess is controlled.	Timing is debated; some treat both at one sitting when the patient is stable.

3.5 Lateral sinus thrombophlebitis

3.5.1 Picture

Inflammation of the sigmoid sinus wall, mural thrombus, then an infected clot. The fever is the clue: high, swinging, *picket-fence* fever with rigors and well periods

in between, like malaria but irregular.

3.5.2 Signs

- **Griesinger's sign:** oedema over the mastoid emissary vein.
- **Tobey-Ayer test:** compressing the internal jugular vein on the healthy side raises CSF pressure, but compressing the affected side does not.
- **Crowe-Beck test:** compressing the healthy jugular vein engorges the retinal veins.
- Papilloedema, more often with a right-sided thrombus as the right sinus is usually dominant.
- Tenderness along the jugular vein; septic emboli to the lungs, bones and joints.

3.5.3 Investigations

Blood culture during a rigor, a smear to exclude malaria, and contrast CT (the *empty delta sign*) or MR venography.

3.5.4 Treatment

Intravenous antibiotics and mastoidectomy with exposure of the sinus. If it is needle-aspirated and yields pus or no blood, it is opened between packs placed above and below, and infected clot is removed; healthy clot is left alone. Anticoagulation is debated. Ligation of the internal jugular vein is kept for septic emboli or clot spreading down the neck.

3.6 Otitic hydrocephalus

3.6.1 Picture

Raised intracranial pressure with normal CSF and normal-sized ventricles, usually after lateral sinus thrombosis that blocks venous outflow. Headache, vomiting, blurred vision, papilloedema and a sixth nerve palsy.

3.6.2 CSF

Pressure over 300 mm of water, with normal contents.

3.6.3 Treatment

Treat the ear and the sinus; lower the pressure with acetazolamide, steroids and mannitol, with repeated lumbar punctures or a shunt if needed. Serial eye examinations are essential, because the danger is blindness from optic atrophy.

4 CHAPTER FOUR Investigations

Image before you puncture.

4.1 Imaging

4.1.1 Contrast CT of the brain and temporal bone

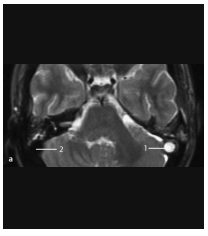
The first study: abscess, empyema, bone erosion and the delta sign.

4.1.2 MRI with contrast and MR venography

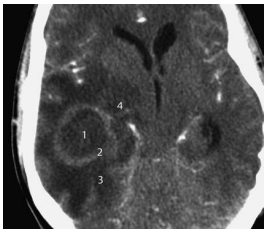
More sensitive for early cerebritis, subdural collections and sinus thrombosis.

IMAGING IN THE APP 5 cases

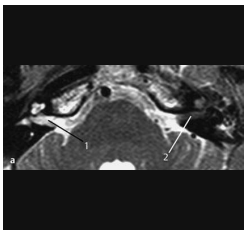
Sinus thrombosis, otogenic abscess, labyrinthitis with meningitis, and petrositis.



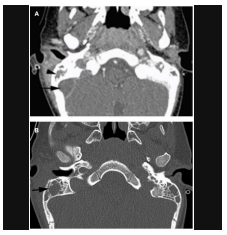
Dural Sinus Thrombosis
Infection & inflammation · 10 images



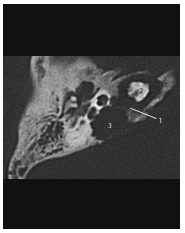
Intracranial Abscess from an Otologic Focus
Infection & inflammation · 1 image



Labyrinthitis and Subclinical Meningitis
Infection & inflammation · 5 images



Otomastoiditis and its complications
Infection & inflammation · 9 images



Petrositis
Infection & inflammation · 3 images

4.2 Lumbar puncture

Only when imaging has excluded mass effect and raised pressure.

CSF BY COMPLICATION

COMPLICATION	PRESSURE	CELLS	SUGAR
Meningitis	Raised	Very high, polymorphs	Low
Brain abscess	Raised	Moderate	Normal

COMPLICATION	PRESSURE	CELLS	SUGAR
Sinus thrombosis	Raised	Normal	Normal
Otitic hydrocephalus	Over 300 mm H2O	Normal	Normal

4.3 Other tests

Blood cultures, full blood count, ear swab, audiogram when possible.

WHICH COMPLICATION?

COMPLICATION	CLUE	NEXT STEP
Meningitis	Neck stiffness	CT, then lumbar puncture
Temporal abscess	Aphasia, field defect	MRI, drain
Cerebellar abscess	Ataxia, nystagmus	MRI, drain
Sinus thrombosis	Picket-fence fever	MR venography
Otitic hydrocephalus	Papilloedema, clear CSF	MRV, lower pressure

5 CHAPTER FIVE Management

Patient first, ear last.

5.1 The sequence

ON CALL: DISCHARGING EAR WITH FEVER AND HEADACHE

1. Airway, breathing, circulation; treat sepsis.
2. Full neurological examination, including fundi and cranial nerves.
3. Contrast CT of brain and temporal bones, then MRI and MRV.
4. Start intravenous antibiotics with CSF penetration and anaerobic cover.
5. Call neurosurgery early.

6 No lumbar puncture until mass effect and raised pressure are excluded.

5.2 Treating the ear

Acute otitis media: myringotomy and grommet. Chronic disease or cholesteatoma: mastoidectomy to remove the source, once the intracranial problem is controlled and the patient is fit.

OPERATIVE GUIDES IN THEATRE 2 operations

Intracranial Complications of Otitis Media
Intracranial complications

Mastoid Surgery
Mastoid source control

5.3 Outcomes

Deafness, epilepsy, focal deficits, loss of vision and death. Rupture of an abscess into the ventricle carries a mortality of about 80%.

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Routes of spread:** erosion, thrombophlebitis, preformed pathways.
- **Stages of brain abscess,** and the danger of the quiet stage.
- **Temporal versus cerebellar abscess signs.**
- **Lateral sinus thrombosis:** picket-fence fever, Griesinger's sign, Tobey-Ayer and Crowe-Beck tests, delta sign.
- **CSF in each complication.**

- **Otitic hydrocephalus:** raised pressure, normal CSF, risk of blindness.

6.2 Common pitfalls

COMMON PITFALLS

- Doing a lumbar puncture before imaging.
- Being reassured by the quiet stage of an abscess.
- Treating swinging fever as malaria without examining the ear.
- Stopping at one complication and missing a second.
- Forgetting a hearing test after meningitis.

Written for this app from Oxford Textbook of Otolaryngology ch. 8, Oxford Specialist Handbook ch. 16, Oxford Handbook of ENT ch. 6, Dhingra ch. 12 and K. J. Lee ch. 22. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

MIDDLE EAR & MASTOID · INFECTION & INFLAMMATION

Dural Sinus Thrombosis

IMAGING FINDINGS

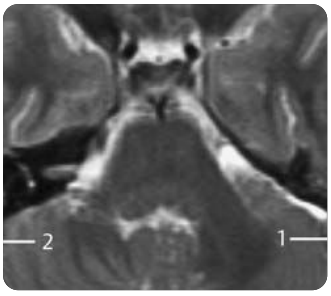
- Filling defect in the sigmoid or transverse sinus on contrast CT or MR venography
- Empty delta sign; dense sinus on non-contrast CT
- Adjacent mastoiditis with bone erosion over the sinus

TOP DIFFERENTIALS

- Arachnoid granulation (round, CSF signal)
- Hypoplastic transverse sinus
- Slow flow artefact

CHECKLIST

- Extension to the jugular vein and other sinuses
- Venous infarction and raised intracranial pressure



MRI, T2-weighted, axial. Otomastoiditis can spread intracranially, resulting in dural sinus thrombophlebitis, which may compromise the sigmoid and…



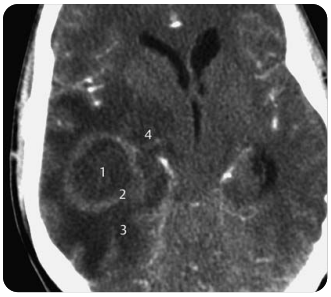
Axial CECT in a patient with postauricular tender mass, headache, and fever reveals postauricular abscess and coalescent…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1042.

MIDDLE EAR & MASTOID · INFECTION & INFLAMMATION

Intracranial Abscess from an Otologic Focus

- Otogenic abscesses lie in the temporal lobe (through the tegmen) or the cerebellum (through Trautmann's triangle or the sigmoid sinus).
- MRI: a ring-enhancing lesion with restricted diffusion in the centre (unlike necrotic tumour) and surrounding oedema.
- Treat with neurosurgical drainage, long-course antibiotics and, once the patient is stable, mastoid surgery to remove the source.

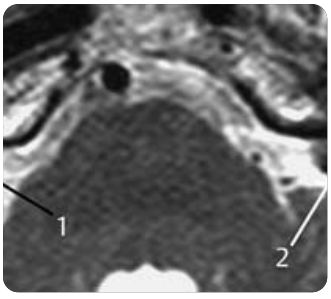


CT with intravenous contrast, axial. Otomastoiditis may spread intracranially resulting in meningitis, encephalitis, and/ or brain abscess....

INNER EAR & IAC · INFECTION & INFLAMMATION

Labyrinthitis and Subclinical Meningitis

- Otoscopy may exclude middle ear infections although CT may be indicated in some cases. MRI will show (retro)cochlear enhancement in cases of (acute) infections.
- Inflammatory involvement of the intralabyrinthine structures may result in fibrosis and ossification. In bilateral cases, this may hamper insertion of a cochlear implant.



MRI, T2-weighted, axial. Note the normal fluid content of the internal auditory canal on the right side (1), whereas on the affected side, the...

MIDDLE EAR & MASTOID · INFECTION & INFLAMMATION

Otomastoiditis and its complications

IMAGING FINDINGS

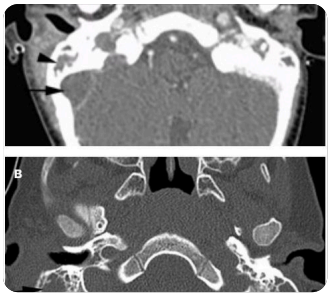
- Opacified middle ear and mastoid
- Coalescence: loss of the mastoid septa and cortical erosion
- Complications: subperiosteal abscess, sigmoid sinus thrombosis, epidural or brain abscess, labyrinthitis

TOP DIFFERENTIALS

- Uncomplicated effusion (septa intact)
- Cholesteatoma
- Langerhans cell histiocytosis or rhabdomyosarcoma in a child

CHECKLIST

- Contrast CT or MRI with venography when complications are suspected
- Check the sigmoid sinus, tegmen, labyrinth and petrous apex



Post-contrast CT shows (a) right coalescent mastoiditis (black arrowhead) and ipsilateral sigmoid sinus thrombosis (black arrow) – bot…



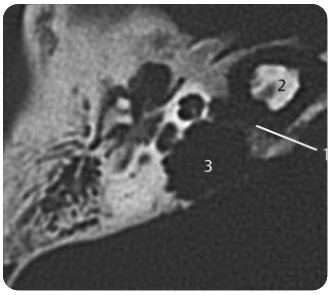
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TEMPORAL BONE & SKULL BASE · INFECTION & INFLAMMATION

Petrositis

- Clinical symptoms may help differentiate between inflammatory disorders, asymptomatic pseudotumors, and other pathology. CT and MRI are complementary. Knowing whether the origin is intradural or extradural and the presence of bony changes help to establish the most likely diagnosis.



CT, axial. The internal auditory canal is seen (1), but is not clearly recognizable. Anteriorly, a rounded lucency in the petrous apex is visible…